

INTELLIGENT TAXI ROUTE OPTIMIZATION USING REINFORCEMENT LEARNING

Optimizing Taxi Routes using **DQN, Dijkstra & A***

SUBMITTED TO - MS.KHUSHI RATTAN(IBM)

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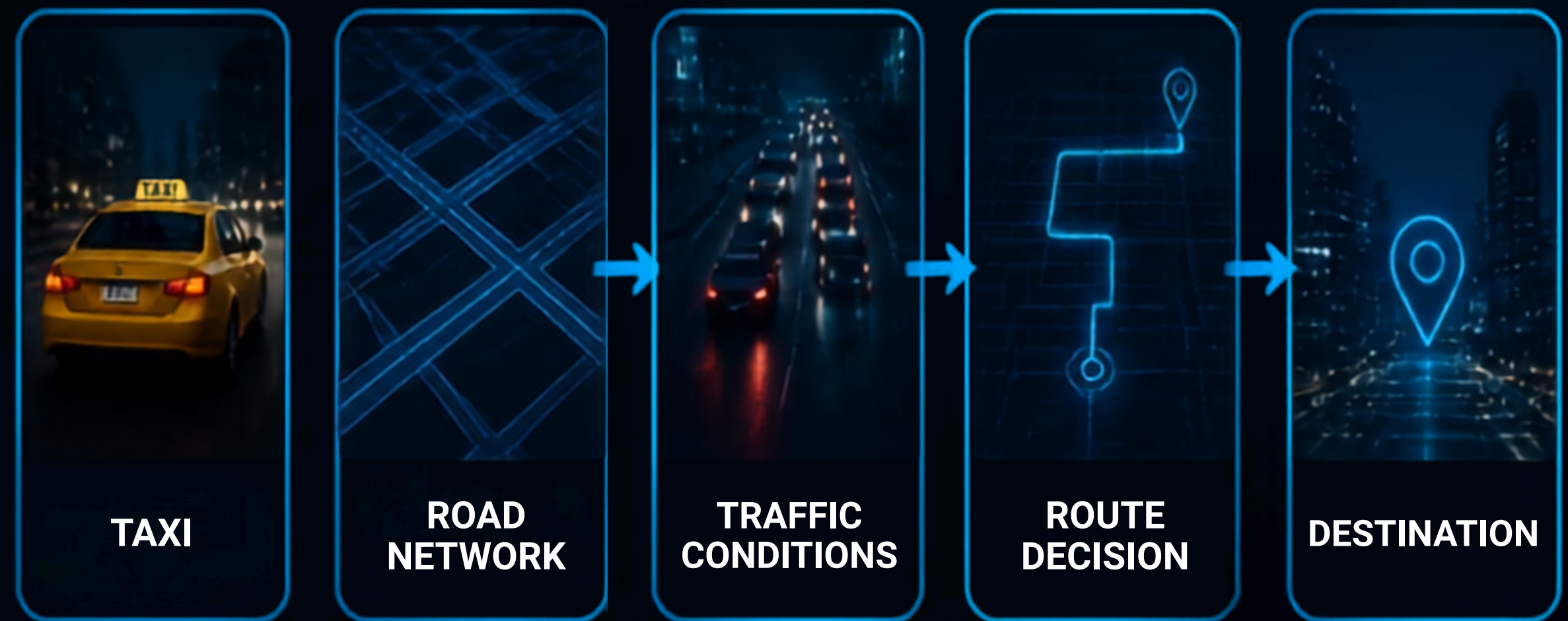
PROBLEM STATEMENT

Taxi routing must balance distance, time, traffic, and operational cost

Traditional shortest-path methods rely on fixed, static costs

Traffic conditions are dynamic and unpredictable

Reinforcement learning enables **adaptive** route decisions through **environment interaction**



OBJECTIVES



Develop an intelligent taxi route optimization system



Compare Dijkstra, A* and DQN algorithms



Incorporate simulated dynamic traffic conditions



Optimize multiple objectives: cost, distance, travel time



Provide comparative route analysis and quantitative evaluation



Build a web-based interactive dashboard

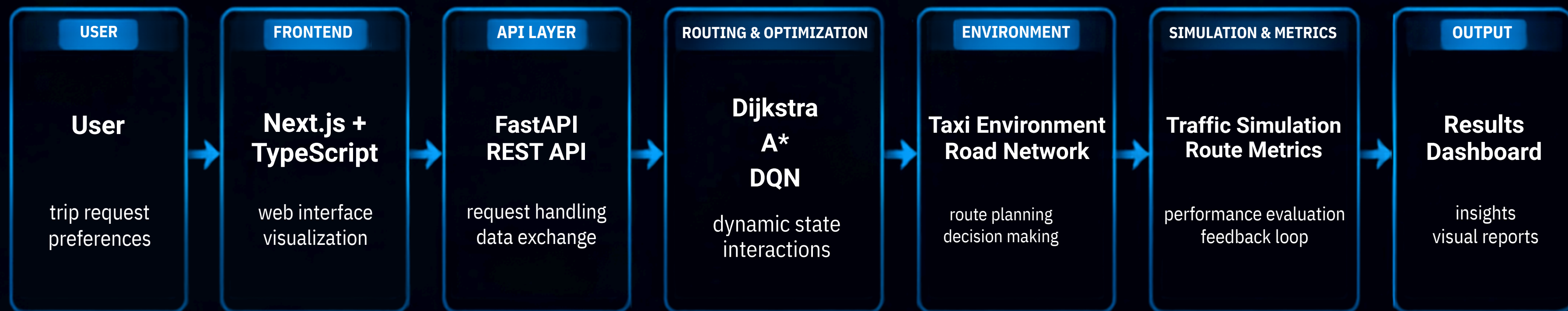


Provide training analytics for the reinforcement learning model

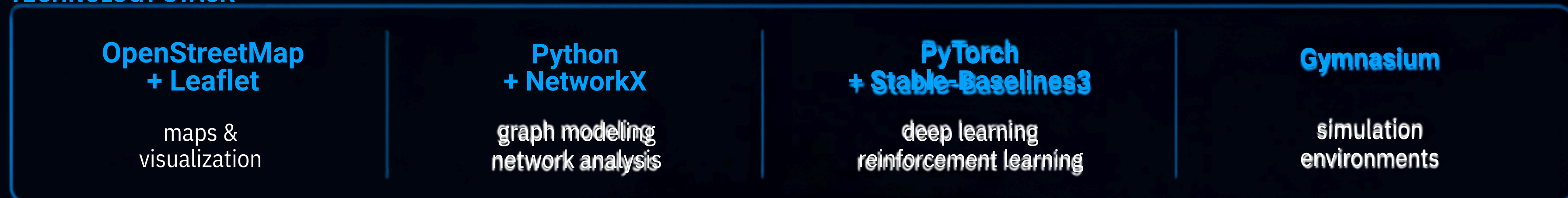


SYSTEM ARCHITECTURE

end-to-end flow of the intelligent taxi route optimization system

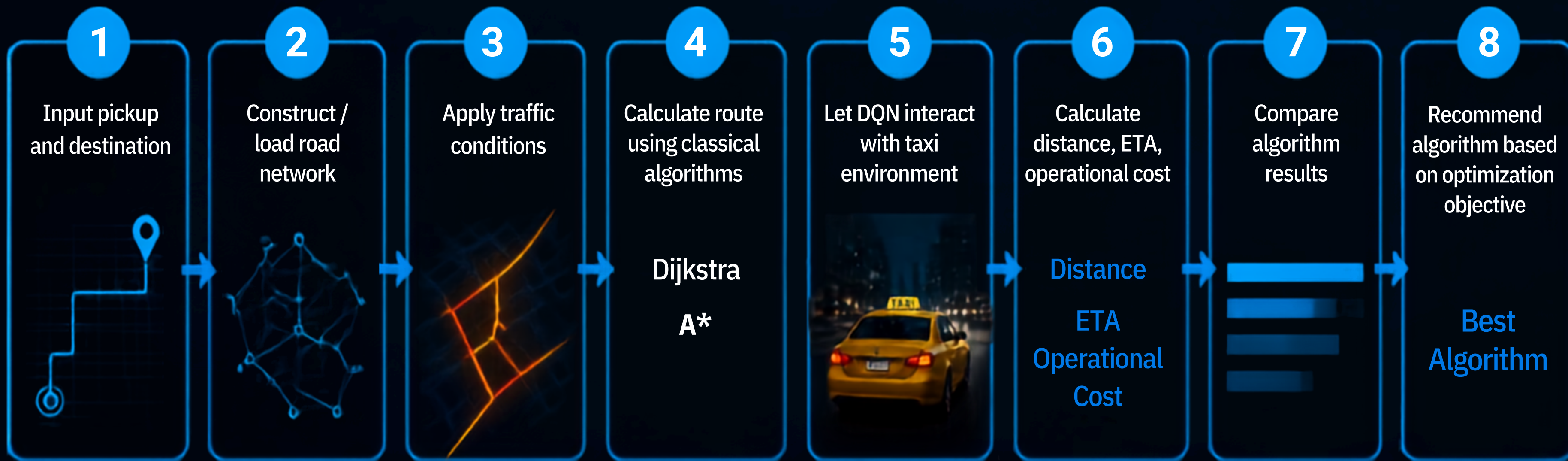


TECHNOLOGY STACK



METHODOLOGY

A stepwise framework that integrates classical routing algorithms with deep reinforcement learning to recommend **optimal taxi routes**.

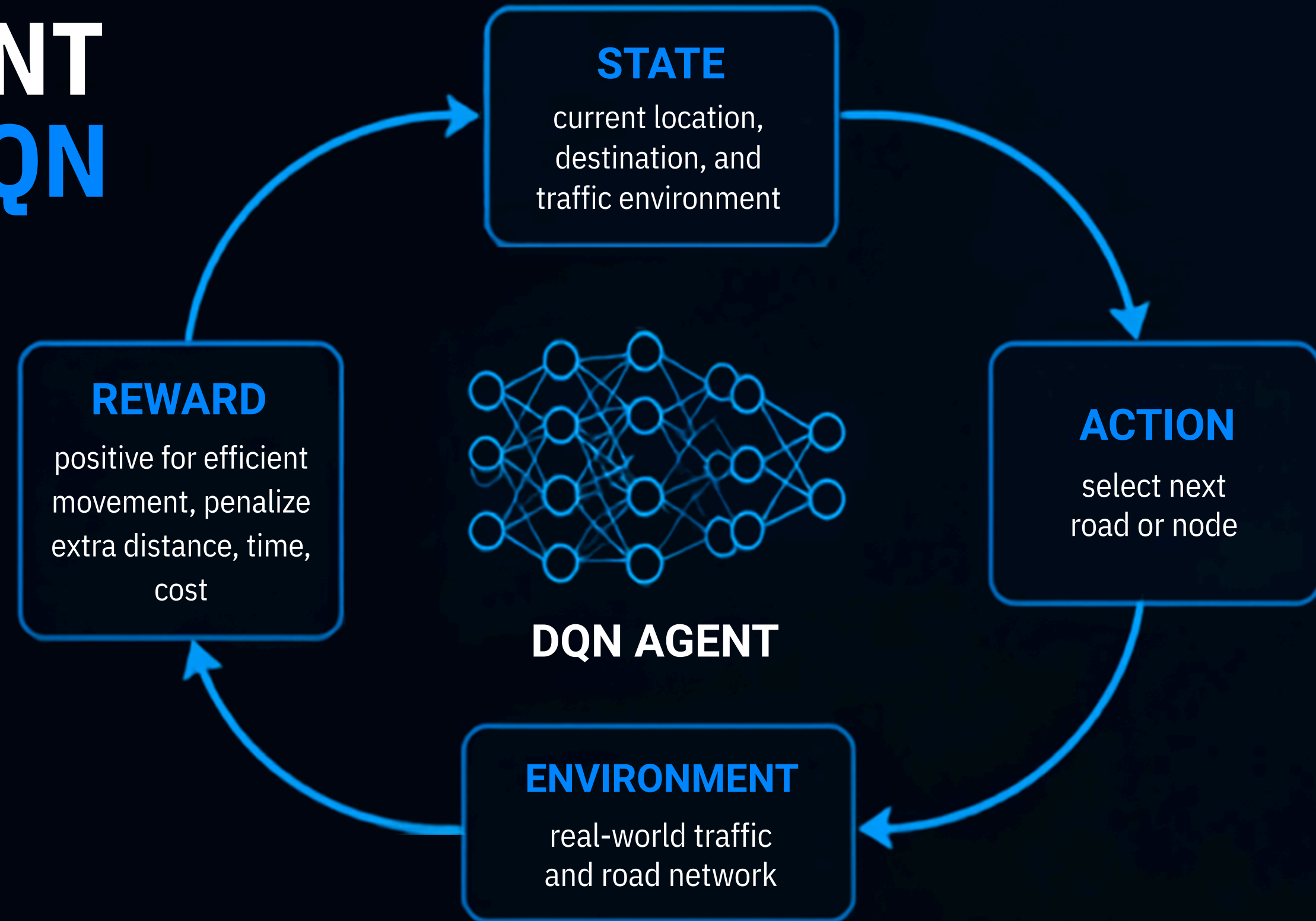


REINFORCEMENT LEARNING & DQN

DQN **learns** to choose the best route through trial and feedback

It observes the current **state**, takes an **action**, and receives a **reward**

The agent **improves** its policy to select routing actions that minimize time, distance, and cost



NEURAL NETWORK

approximates Q-values for state-action pairs

EXPERIENCE REPLAY

stores past experiences and samples for stable learning

TARGET NETWORK

uses delayed updates to stabilize training

Q-VALUE

expected return of an action from a given state



CLASSICAL BASELINES



APPROACH	TYPE	MAIN PRINCIPLE	STRENGTH	LIMITATION
Dijkstra	Exact shortest-path	No heuristic needed	Reliable baseline	Computationally intensive on large graphs
A*	Heuristic guided search	Uses heuristic to prune search	Efficient with good heuristic	Depends on heuristic quality
DQN	Learning-based	Learns routing policy by interaction	Adapts to dynamic environments	Requires training, less predictable

TECHNOLOGY STACK & IMPLEMENTATION



FRONTEND

Next.js
TypeScript
Tailwind CSS
Leaflet



BACKEND

FastAPI
Python
REST API



AI / ML

PyTorch
Stable-Baselines3
Gymnasium



ROUTING

NetworkX
Dijkstra
A*



MAPS & DATA

OpenStreetMap
Synthetic taxi-trip data
Traffic simulation



DEPLOYMENT

Vercel
Render
GitHub



RESULTS & DASHBOARD

Live system dashboard showing optimized route, key outputs, and algorithm comparison in a simulated environment.



CONCLUSION & FUTURE SCOPE

We built an **intelligent taxi routing** platform

Classical algorithms set a **strong baseline**

DQN brings **reinforcement learning** to routing

End-to-end system with **simulation, AI, and dashboard**

A modular architecture ready for **future innovation**

THANK YOU | **QUESTIONS?**

